Population Attributable Fraction of Obesity for Hypercapnic Respiratory Failure

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1 Obesity and hypercapnic respiratory failure analysis

1.1 Section 0. Setup

1.1.1 Load packages

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.4     v readr     2.1.5
v forcats   1.0.0     v stringr   1.5.1
v ggplot2   3.5.2     v tibble    3.2.1
v lubridate 1.9.4     v tidyr     1.3.1
v purrr     1.0.4
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(haven)
library(ggplot2)
library(modelsummary)
library(here)
```

```
here() starts at /Users/blocke/Box Sync/Residency Personal Files/Scholarly Work/Locke Research Projects/obes-pop-attrib-fract-hypercap
```

```
rm(list = ls())
# TODO: renv::snapshot()
```

1.1.2 Define file paths and constants

```
data_path <- here::here("full_db.rds")

co2_abg_cutoff <- 45
co2_vbg_cutoff <- 50
ph_cutoff <- 7.35
icd_hrf_vars <- c("has_j9612", "has_j9622", "has_j9692", "has_j9602")

coef_map_common <- c(
  "curr_bmi_num" = "BMI (per 1 kg/m^2)",
  "age_at_encounter" = "Age (per year)",
  "sexMale" = "Male",
  "ethnicityHispanic or Latino" = "Hispanic/Latino",
  "ethnicityUnknown" = "Ethnicity: Unknown"
)
```

1.2 Section 1. Data ingest and harmonization

1.2.1 readRDS()

```
stata_data <- readRDS(data_path)
```

1.2.2 Standardize types (labelled -> numeric, factors)

```
# Verify before conversion
table(stata_data$niv_proc, useNA = "ifany")
```

```
                                    No NIV
                                    799385
Received NIV (not including nightly CPAP)
                                     34091
```

```
# Convert labelled variables to numeric properly
stata_data <- stata_data %>%
  mutate(
    niv_proc = case_when(
      as.character(as_factor(niv_proc)) == "Received NIV (not including nightly CPAP)" ~ 1,
      as.character(as_factor(niv_proc)) == "No NIV" ~ 0,
      TRUE ~ NA_real_
    ),
    imv_proc = case_when(
      as.character(as_factor(imv_proc)) == "Received IMV" ~ 1,
      as.character(as_factor(imv_proc)) == "No IMV" ~ 0,
      TRUE ~ NA_real_
    ),
    highest_paco2 = as.numeric(highest_paco2),
    abg_ph = as.numeric(abg_ph),
    vbg_co2 = as.numeric(vbg_co2),
    vbg_ph = as.numeric(vbg_ph)
  )

# Verify after conversion
table(stata_data$niv_proc, useNA = "ifany")
```

```
     0      1
799385  34091
```

```r
stopifnot(all(is.na(stata_data$niv_proc) | stata_data$niv_proc %in% c(0, 1)))
stopifnot(any(stata_data$niv_proc == 0, na.rm = TRUE))
stopifnot(any(stata_data$niv_proc == 1, na.rm = TRUE))

stopifnot(all(is.na(stata_data$imv_proc) | stata_data$imv_proc %in% c(0, 1)))
stopifnot(any(stata_data$imv_proc == 0, na.rm = TRUE))
stopifnot(any(stata_data$imv_proc == 1, na.rm = TRUE))

any_vent <- (stata_data$niv_proc == 1 | stata_data$imv_proc == 1)
stopifnot(any(any_vent == TRUE, na.rm = TRUE))
stopifnot(any(any_vent == FALSE, na.rm = TRUE))
```

1.2.3 Quick sanity checks (row counts, missingness)

```r
nrow(stata_data)
```

```
[1] 833476
```

TODO: Add missingness checks.

1.3 Section 2. Variable derivations

1.3.1 BMI parsing + BMI categories

```r
bmi_raw <- stata_data %>%
  transmute(curr_bmi_raw = na_if(str_trim(as.character(zap_labels(curr_bmi))), "n/a"))

non_numeric_bmi <- bmi_raw %>%
```

```
  filter(!is.na(curr_bmi_raw), !str_detect(curr_bmi_raw, "^\\d+(\\.\\d+)?$"))

non_numeric_bmi %>%
  count(curr_bmi_raw, sort = TRUE)
```

curr_bmi_raw	n

```
non_numeric_bmi %>%
  summarise(n_failed_parse = n())
```

n_failed_parse
0

```
stata_data <- stata_data %>%
  mutate(
    curr_bmi_num = if_else(
      str_detect(str_trim(as.character(zap_labels(curr_bmi))), "^\\d+(\\.\\d+)?$"),
      as.numeric(str_trim(as.character(zap_labels(curr_bmi)))),
      NA_real_
    )
  )
```

```
stata_data <- stata_data %>%
  mutate(
    bmi_category = case_when(
      is.na(curr_bmi_num) ~ NA_character_,
      curr_bmi_num < 18.5 ~ "Underweight",
      curr_bmi_num < 25   ~ "Normal weight",
      curr_bmi_num < 30   ~ "Overweight",
      curr_bmi_num < 35   ~ "Obesity I",
      curr_bmi_num < 40   ~ "Obesity II",
      curr_bmi_num >= 40  ~ "Obesity III"
```

```
    )
  )
```

1.3.2 HRF definition v1 (gases + ventilation)

Below is my first (probably most stringent) iteration of hypercapnic respiratory failure: co2 greater than 45 if abg, 50 if vbg, ph <7.35 - they have to have either an abg or a vbg showing this, AND either NIV or IMV or both. I've written it so that ABG values and VBG values cannot mix

```
stata_data <- stata_data %>%
  mutate(
    gas_source = case_when(
      !is.na(highest_paco2) & !is.na(abg_ph) ~ "ABG",
      TRUE ~ "VBG"
    )
  )
```

```
stata_data <- stata_data %>%
  mutate(
    hrf_hypercapnic = case_when(
      # ABG-based HRF
      (niv_proc == 1 | imv_proc == 1) &
        gas_source == "ABG" &
        highest_paco2 >= co2_abg_cutoff &
        abg_ph <= ph_cutoff ~ 1,

      # VBG-based HRF (only when ABG incomplete)
      (niv_proc == 1 | imv_proc == 1) &
        gas_source == "VBG" &
        !is.na(vbg_co2) & !is.na(vbg_ph) &
        vbg_co2 >= co2_vbg_cutoff &
        vbg_ph <= ph_cutoff ~ 1,

      TRUE ~ 0
    ),
```

```
    hrf_hypercapnic = factor(
      hrf_hypercapnic,
      levels = c(0, 1),
      labels = c("No", "Yes")
    )
  )
```

1.3.3 HRF definition v2 (labs + ICD) and v2_strict

Regression done with new definition of hypercapnic respiratory failure - pH and CO2 definition as above, except now including (required) ICD codes instead of ventilation (I only used 4 - the first four that are listed in database):

```
stata_data <- stata_data %>%
  mutate(
    # Lab-based hypercapnic respiratory failure (ABG or VBG)
    hrf_hypercapnic_labs_v2 = case_when(
      !is.na(highest_paco2) & !is.na(abg_ph) &
       highest_paco2 >= co2_abg_cutoff & abg_ph < ph_cutoff ~ 1, # BL note - fixed this 50->45

      !is.na(vbg_co2) & !is.na(vbg_ph) &
        vbg_co2 >= co2_vbg_cutoff & vbg_ph < ph_cutoff ~ 1,

      TRUE ~ 0
    ),

    # ICD-based hypercapnic respiratory failure
    hrf_hypercapnic_icd_v2 = ifelse(
      rowSums(across(all_of(icd_hrf_vars)) == 1, na.rm = TRUE) > 0,
      1, 0
    ),

    # Final combined HRF definition (labs OR ICD)
    hrf_hypercapnic_v2 = ifelse(
      hrf_hypercapnic_labs_v2 == 1 |
      hrf_hypercapnic_icd_v2 == 1,
```

```
      1, 0
    )
  )
```

```
stata_data <- stata_data %>%
  mutate(
    hrf_hypercapnic_v2_strict = ifelse(
      hrf_hypercapnic_labs_v2 == 1 &
      hrf_hypercapnic_icd_v2  == 1,
      1, 0
    )
  )
```

1.4 Section 3. Cohort construction and attrition

1.4.1 Primary analysis cohort (HRF v1)

```
stata_hrf_analysis <- stata_data %>%
  filter(!is.na(curr_bmi_num),
         (!is.na(highest_paco2) & !is.na(abg_ph)) |
         (!is.na(vbg_co2) & !is.na(vbg_ph)))
```

1.4.2 Strict v2 cohort (sensitivity)

```
stata_hrf_strict_analysis <- stata_data %>%
  mutate(
    has_gas = (!is.na(highest_paco2) & !is.na(abg_ph)) |
              (!is.na(vbg_co2) & !is.na(vbg_ph))
  ) %>%
  filter(
    # Keep strict HRF cases
    hrf_hypercapnic_v2_strict == 1 |
```

```
    # OR clean controls with labs and no HRF by any definition
    (
      has_gas == TRUE &
      hrf_hypercapnic_labs_v2 == 0 &
      hrf_hypercapnic_icd_v2  == 0
    )
  )
```

TODO: Add attrition table with inclusion/exclusion counts.

1.5 Section 4. Descriptive statistics

1.5.1 Distribution of BMI in the analytic cohort

```
table(stata_data$hrf_hypercapnic, useNA = "ifany")
```

```
    No    Yes
807702  25774
```

```
summary(stata_data$curr_bmi_num)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
  10.09   26.31   33.34   33.64   41.20   50.00  450715
```

```
table(stata_data$bmi_category, useNA = "ifany")
```

```
Normal weight     Obesity I    Obesity II   Obesity III    Overweight
        65912         58317         44022        130477         74499
  Underweight          <NA>
         9534        450715
```

[BL - note high rates of BMI missing data]

1.5.2 HRF prevalence by BMI category (v1)

Counts (bottom right is amount of patients meeting criteria above, adds up to full dataset)

```
table(
  hrf = stata_data$hrf_hypercapnic,
  any_vent = (stata_data$niv_proc == 1 | stata_data$imv_proc == 1)
)
```

```
     any_vent
hrf    FALSE   TRUE
  No  747798  59904
  Yes      0  25774
```

1.5.3 Counts for both HRF as defined above AND current BMI above 30

```
stata_data %>%
  filter(
    hrf_hypercapnic == "Yes",
    curr_bmi_num >= 30
  ) %>%
  summarise(
    n_patients = n()
  )
```

n_patients
4099

Counts for HRF and BMI by category

```r
stata_data %>%
  filter(
    hrf_hypercapnic == "Yes",
    !is.na(curr_bmi_num)
  ) %>%
  mutate(
    bmi_category = case_when(
      curr_bmi_num < 18.5 ~ "Underweight",
      curr_bmi_num < 25   ~ "Normal weight",
      curr_bmi_num < 30   ~ "Overweight",
      curr_bmi_num < 35   ~ "Obesity I",
      curr_bmi_num < 40   ~ "Obesity II",
      curr_bmi_num >= 40  ~ "Obesity III"
    )
  ) %>%
  count(bmi_category) %>%
  mutate(pct = round(100 * n / sum(n), 1))
```

bmi_category	n	pct
Normal weight	2311	25.1
Obesity I	1766	19.2
Obesity II	1131	12.3
Obesity III	1202	13.1
Overweight	2287	24.9
Underweight	503	5.5

1.5.4 Baseline characteristics table

☐ TODO

1.5.5 Are proportions of hypercapnic respiratory failure (v1) different across BMI categories?

Just out of curiosity ran Chi square analysis below

```r
bmi_hrf_subset <- stata_hrf_analysis %>%
  filter(!is.na(bmi_category))
```

```r
bmi_hrf_summary <- bmi_hrf_subset %>%
  group_by(bmi_category) %>%
  summarise(
    n_total = n(),
    n_hrf = sum(hrf_hypercapnic == "Yes"),
    n_no_hrf = sum(hrf_hypercapnic == "No"),
    pct_hrf = round(100 * n_hrf / n_total, 1),
    pct_no_hrf = round(100 * n_no_hrf / n_total, 1)
  ) %>%
  arrange(factor(bmi_category, levels = c("Underweight", "Normal weight", "Overweight",
                                          "Obesity I", "Obesity II", "Obesity III")))

chisq_test <- chisq.test(table(bmi_hrf_subset$bmi_category, bmi_hrf_subset$hrf_hypercapnic))

bmi_hrf_summary
```

bmi_category	n_total	n_hrf	n_no_hrf	pct_hrf	pct_no_hrf
Underweight	5061	503	4558	9.9	90.1
Normal weight	31060	2311	28749	7.4	92.6
Overweight	31756	2287	29469	7.2	92.8
Obesity I	21692	1766	19926	8.1	91.9
Obesity II	12223	1131	11092	9.3	90.7
Obesity III	10217	1202	9015	11.8	88.2

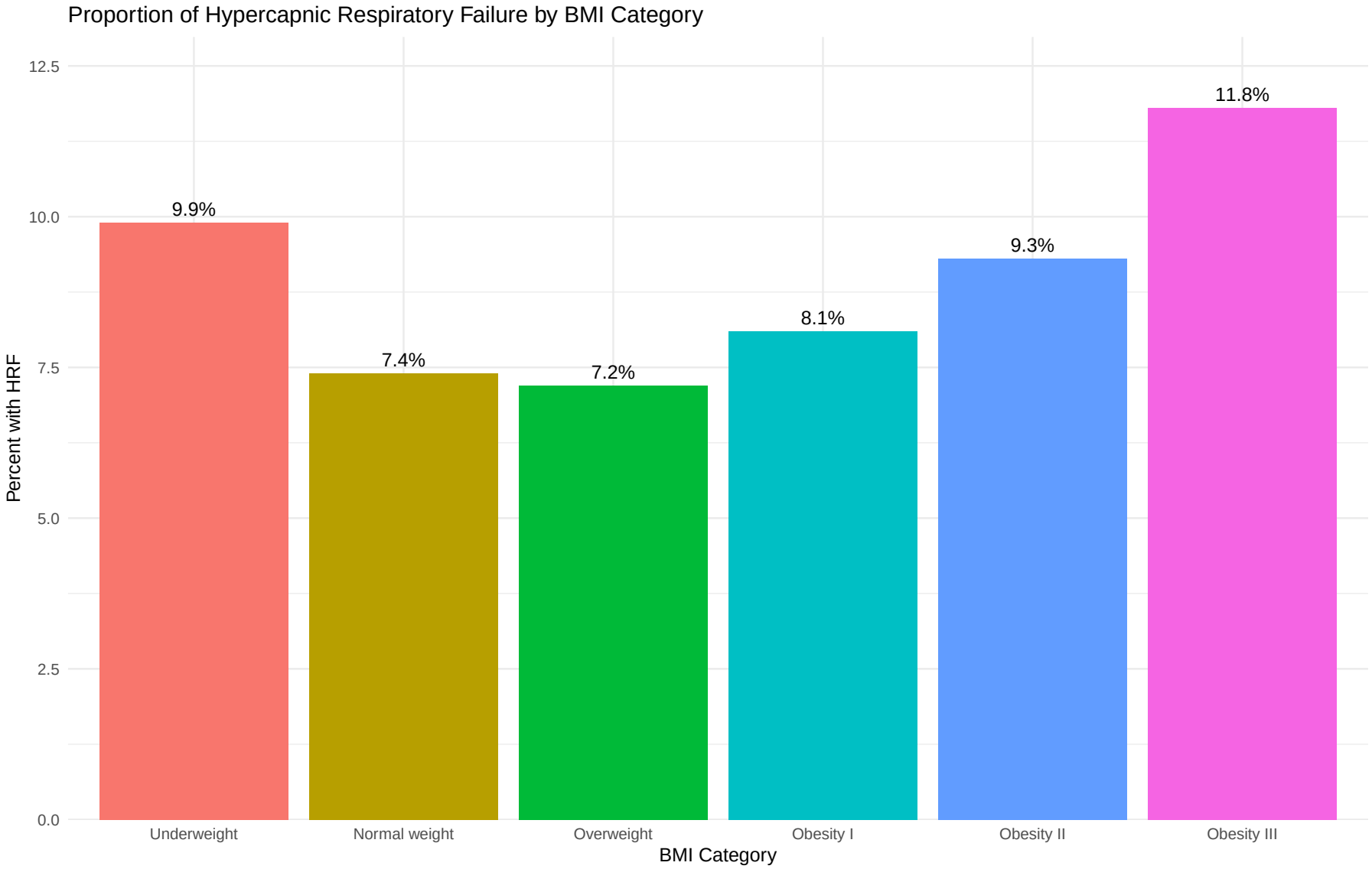
```r
chisq_test$p.value
```

```
[1] 1.243203e-57
```

Visualization:

```r
# Make sure bmi_category is ordered
bmi_hrf_summary <- bmi_hrf_summary %>%
  mutate(bmi_category = factor(bmi_category,
                               levels = c("Underweight", "Normal weight", "Overweight",
                                          "Obesity I", "Obesity II", "Obesity III")))

# Plot proportion of HRF by BMI category
ggplot(bmi_hrf_summary, aes(x = bmi_category, y = pct_hrf, fill = bmi_category)) +
  geom_col() +
  geom_text(aes(label = paste0(pct_hrf, "%")), vjust = -0.5) +
  scale_y_continuous(expand = expansion(mult = c(0, 0.1))) +
  labs(
    title = "Proportion of Hypercapnic Respiratory Failure by BMI Category",
    x = "BMI Category",
    y = "Percent with HRF"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```


Proportion of Hypercapnic Respiratory Failure by BMI Category
Percent with HRF
12.5
10.0
7.5
5.0
2.5
0.0
9.9%
7.4%
7.2%
8.1%
9.3%
11.8%
Underweight
Normal weight
Overweight
Obesity I
Obesity II
Obesity III
BMI Category

1.6 Section 5. Primary inferential model

1.6.1 Unadjusted (aka simple) regression for outcome 1

Below is regression, starting with BMI (the degrees of freedom is 112k which seems appropriate - includes all patients with a BMI and labs/vent indicating NO HRF, plus the patients with BMI and hypercapnic respiratory failure as defined above)

```
logit_bmi_hrf <- glm(hrf_hypercapnic ~ curr_bmi_num,
data = stata_hrf_analysis,
family = binomial)

summary(logit_bmi_hrf)
```

```
Call:
glm(formula = hrf_hypercapnic ~ curr_bmi_num, family = binomial,
    data = stata_hrf_analysis)

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  -2.92776    0.04482  -65.33   <2e-16 ***
curr_bmi_num  0.01750    0.00146   11.98   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 63611  on 112008  degrees of freedom
Residual deviance: 63470  on 112007  degrees of freedom
AIC: 63474

Number of Fisher Scoring iterations: 5
```

OR shows small increase in risk with BMI, though is statistically significant

[BL Edit - though keep in mind, Odds increased by .02 for each BMI point (e.g. 10 = 20% increase in odds)]

1.6.2 Adjusted, multivariable regression for outcome 1

Regression done again adjusted for variables NOT influenced by BMI (age, sex, ethnicity included - didn't include location as it appears to be the same for all entries)

```
baseline_model <- glm(
hrf_hypercapnic ~ curr_bmi_num + age_at_encounter + sex + ethnicity,
data = stata_hrf_analysis,
family = binomial
)

summary(baseline_model)
```

```
Call:
glm(formula = hrf_hypercapnic ~ curr_bmi_num + age_at_encounter +
    sex + ethnicity, family = binomial, data = stata_hrf_analysis)

Coefficients:
                             Estimate Std. Error z value Pr(>|z|)
(Intercept)                -3.3143678  0.0653822 -50.692  < 2e-16 ***
curr_bmi_num                0.0194994  0.0014798  13.177  < 2e-16 ***
age_at_encounter            0.0028333  0.0006493   4.363 1.28e-05 ***
sexMale                     0.2388572  0.0222256  10.747  < 2e-16 ***
ethnicityHispanic or Latino -0.0444911  0.0432380  -1.029    0.303
ethnicityUnknown            0.1622532  0.0302736   5.360 8.34e-08 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 63611  on 112008  degrees of freedom
Residual deviance: 63302  on 112003  degrees of freedom
AIC: 63314

Number of Fisher Scoring iterations: 5
```

	HRF v1: unadjusted	HRF v1: adjusted
BMI (per 1 kg/m^2)	1.018	1.020
	[1.015, 1.021]	[1.017, 1.023]
Age (per year)		1.003
		[1.002, 1.004]
Male		1.270
		[1.216, 1.326]
Hispanic/Latino		0.956
		[0.878, 1.040]
Ethnicity: Unknown		1.176
		[1.108, 1.248]
Num.Obs.	112 009	112 009
AIC	63 473.9	63 313.9
BIC	63 493.2	63 371.7
Log.Lik.	−31 734.972	−31 650.965

```
mods_v1 <- list(
  "HRF v1: unadjusted" = logit_bmi_hrf,
  "HRF v1: adjusted"   = baseline_model
)

modelsummary(
  mods_v1,
  exponentiate = TRUE,
  statistic    = "conf.int",
  coef_map = coef_map_common,
  gof_omit = "RMSE|F|Adj|R2"
)
```

Association appears to be stronger once adjusted

1.7 Section 6. Sensitivity analyses

1.7.1 Counts for strict HRF definition (v2_strict)

Number of patients meeting strict definition

```
table(stata_data$hrf_hypercapnic_v2_strict, useNA = "ifany")
```

```
     0      1
817895  15581
```

1.7.2 Unadjusted Logistic Regression for Outcome 2

```
model_strict_bmi_unadj <- glm(
  hrf_hypercapnic_v2_strict ~ curr_bmi_num,
  family = binomial,
  data = stata_hrf_strict_analysis
)

summary(model_strict_bmi_unadj)
```

```
Call:
glm(formula = hrf_hypercapnic_v2_strict ~ curr_bmi_num, family = binomial,
    data = stata_hrf_strict_analysis)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  -3.449156   0.059003  -58.46   <2e-16 ***
curr_bmi_num  0.023881   0.001909   12.51   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 39644  on 87024  degrees of freedom
Residual deviance: 39491  on 87023  degrees of freedom
  (142832 observations deleted due to missingness)
AIC: 39495

Number of Fisher Scoring iterations: 5
```

1.7.3 Adjusted, Multivariable Regression for Outcome 2

Then adjusted for age, sex, ethnicity:

```r
baseline_model_strict <- glm(
  hrf_hypercapnic_v2_strict ~
    curr_bmi_num +
    age_at_encounter +
    sex +
    ethnicity,
  data = stata_hrf_strict_analysis,
  family = binomial
)

summary(baseline_model_strict)
```

```
Call:
glm(formula = hrf_hypercapnic_v2_strict ~ curr_bmi_num + age_at_encounter +
    sex + ethnicity, family = binomial, data = stata_hrf_strict_analysis)

Coefficients:
                            Estimate Std. Error z value Pr(>|z|)
(Intercept)                -4.7689449  0.0935568 -50.974  < 2e-16 ***
curr_bmi_num                0.0277918  0.0019502  14.251  < 2e-16 ***
```

```
age_at_encounter             0.0194879  0.0009325  20.900  < 2e-16 ***
sexMale                     -0.0351480  0.0287347  -1.223  0.22126
ethnicityHispanic or Latino -0.1912798  0.0616342  -3.103  0.00191 **
ethnicityUnknown             0.0537704  0.0416200   1.292  0.19638
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 39644  on 87024  degrees of freedom
Residual deviance: 38980  on 87019  degrees of freedom
  (142832 observations deleted due to missingness)
AIC: 38992

Number of Fisher Scoring iterations: 6
```

```r
mods_v2 <- list(
  "HRF v2 strict: unadjusted" = model_strict_bmi_unadj,
  "HRF v2 strict: adjusted"   = baseline_model_strict
)

modelsummary(
  mods_v2,
  exponentiate = TRUE,
  statistic    = "conf.int",
  coef_map = coef_map_common,
  gof_omit = "RMSE|F|Adj|R2"
)
```

```r
mods_all <- list(
  "HRF v1 (primary)"            = baseline_model,
  "HRF v2_strict (sensitivity)" = baseline_model_strict
)

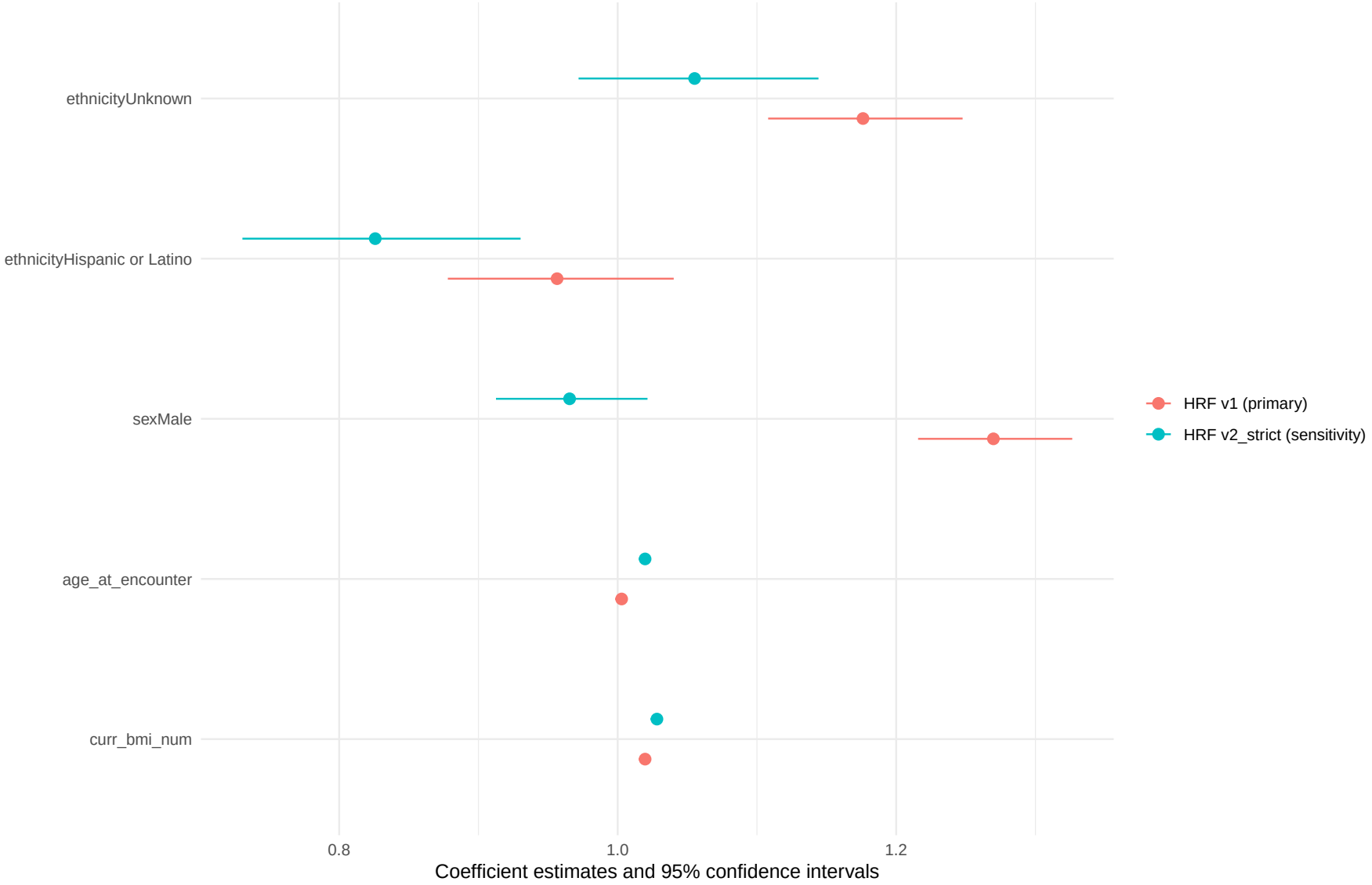
modelsummary(
  mods_all,
```

	HRF v2 strict: unadjusted	HRF v2 strict: adjusted
BMI (per 1 kg/m^2)	1.024	1.028
	[1.020, 1.028]	[1.024, 1.032]
Age (per year)		1.020
		[1.018, 1.022]
Male		0.965
		[0.913, 1.021]
Hispanic/Latino		0.826
		[0.731, 0.930]
Ethnicity: Unknown		1.055
		[0.972, 1.144]
Num.Obs.	87 025	87 025
AIC	39 495.0	38 992.1
BIC	39 513.7	39 048.3
Log.Lik.	−19 745.497	−19 490.036

	HRF v1 (primary)	HRF v2_strict (sensitivity)
BMI (per 1 kg/m^2)	1.020	1.028
	[1.017, 1.023]	[1.024, 1.032]
Age (per year)	1.003	1.020
	[1.002, 1.004]	[1.018, 1.022]
Male	1.270	0.965
	[1.216, 1.326]	[0.913, 1.021]
Hispanic/Latino	0.956	0.826
	[0.878, 1.040]	[0.731, 0.930]
Ethnicity: Unknown	1.176	1.055
	[1.108, 1.248]	[0.972, 1.144]
Num.Obs.	112 009	87 025
AIC	63 313.9	38 992.1
BIC	63 371.7	39 048.3
Log.Lik.	−31 650.965	−19 490.036

```
  exponentiate = TRUE,
  statistic    = "conf.int",
  coef_map = coef_map_common,
  gof_omit = "RMSE|F|Adj|R2"
)
```

```
modelplot(
  mods_all,
  exponentiate = TRUE,
  coef_omit = "(Intercept)"
)
```


ethnicityUnknown
ethnicityHispanic or Latino
sexMale
age_at_encounter
curr_bmi_num
0.8
1.0
1.2
Coefficient estimates and 95% confidence intervals
HRF v1 (primary)
HRF v2_strict (sensitivity)

1.7.4 Are proportions of hypercapnic respiratory failure (v2) different across BMI categories?

```
# Subset to patients with BMI categories
bmi_hrf_strict_subset <- stata_hrf_strict_analysis %>%
  filter(!is.na(bmi_category))

# Summary table by BMI category
bmi_hrf_strict_summary <- bmi_hrf_strict_subset %>%
  group_by(bmi_category) %>%
  summarise(
    n_total = n(),
    n_hrf = sum(hrf_hypercapnic_v2_strict == 1),
    n_no_hrf = sum(hrf_hypercapnic_v2_strict == 0),
    pct_hrf = round(100 * n_hrf / n_total, 1),
    pct_no_hrf = round(100 * n_no_hrf / n_total, 1)
  ) %>%
  arrange(factor(bmi_category, levels = c("Underweight", "Normal weight", "Overweight",
                                          "Obesity I", "Obesity II", "Obesity III")))

# Chi-square test
chisq_test_strict <- chisq.test(table(
  bmi_hrf_strict_subset$bmi_category,
  bmi_hrf_strict_subset$hrf_hypercapnic_v2_strict
))

# View summary table and p-value
bmi_hrf_strict_summary
```

bmi_category	n_total	n_hrf	n_no_hrf	pct_hrf	pct_no_hrf
Underweight	3866	377	3489	9.8	90.2
Normal weight	24496	1301	23195	5.3	94.7
Overweight	25060	1165	23895	4.6	95.4
Obesity I	16721	906	15815	5.4	94.6
Obesity II	9273	666	8607	7.2	92.8

bmi_category	n_total	n_hrf	n_no_hrf	pct_hrf	pct_no_hrf
Obesity III	7609	832	6777	10.9	89.1

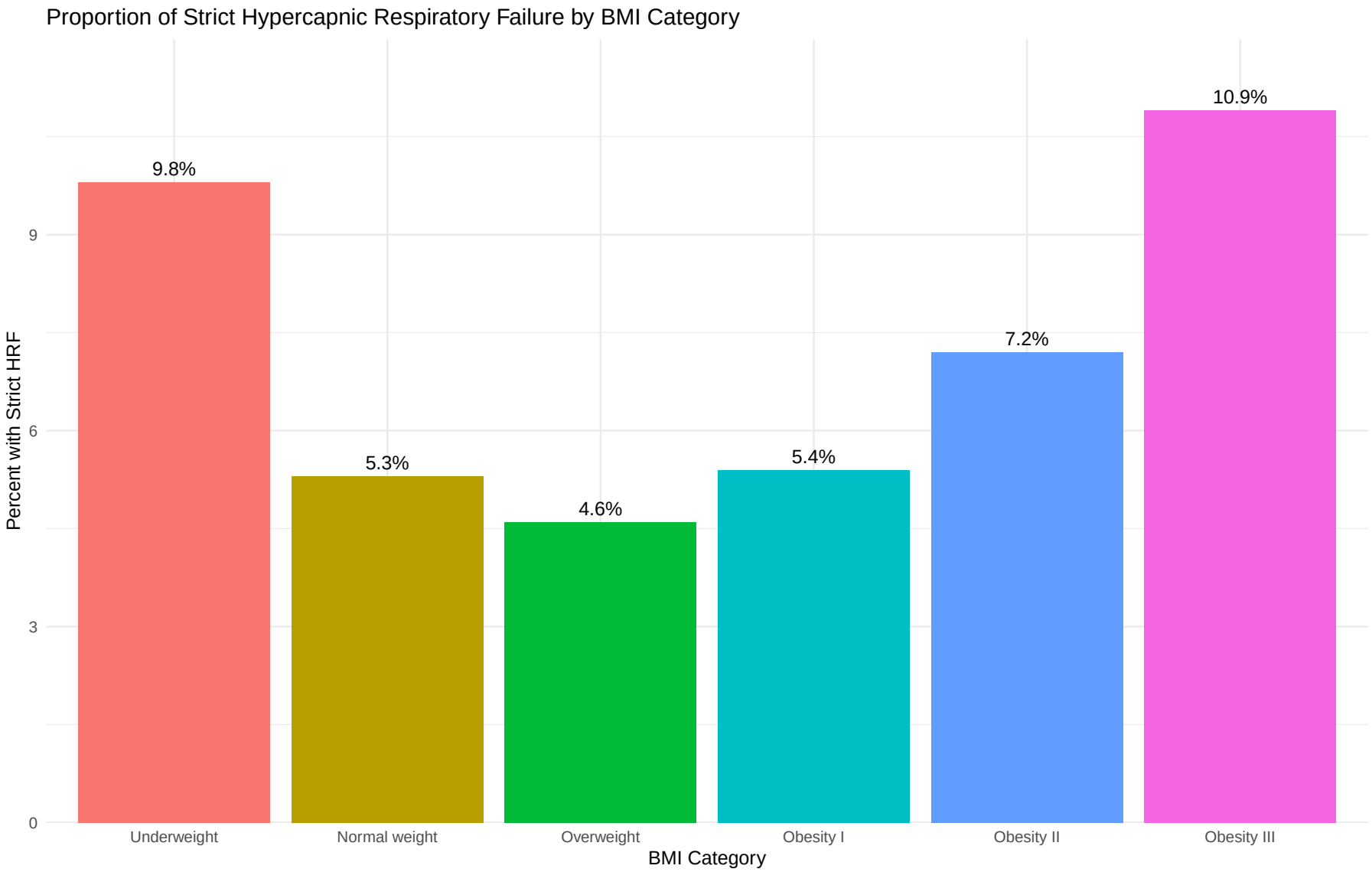
```
chisq_test_strict$p.value
```

```
[1] 3.879776e-118
```

and visualization

```
# Make sure bmi_category is ordered
bmi_hrf_strict_summary <- bmi_hrf_strict_summary %>%
  mutate(bmi_category = factor(bmi_category,
                               levels = c("Underweight", "Normal weight", "Overweight",
                                          "Obesity I", "Obesity II", "Obesity III")))

# Plot proportion of strict HRF by BMI category
ggplot(bmi_hrf_strict_summary, aes(x = bmi_category, y = pct_hrf, fill = bmi_category)) +
  geom_col() +
  geom_text(aes(label = paste0(pct_hrf, "%")), vjust = -0.5) +
  scale_y_continuous(expand = expansion(mult = c(0, 0.1))) +
  labs(
    title = "Proportion of Strict Hypercapnic Respiratory Failure by BMI Category",
    x = "BMI Category",
    y = "Percent with Strict HRF"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```



1.8 Section 7. Population attributable fraction

TODO

1.9 Appendix

```
sessionInfo()
```

```
R version 4.5.1 (2025-06-13)
Platform: aarch64-apple-darwin20
Running under: macOS Tahoe 26.2

Matrix products: default
BLAS:   /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRblas.0.dylib
LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib;  LAPACK version 3.12.1

locale:
[1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8

time zone: America/Denver
tzcode source: internal

attached base packages:
[1] stats     graphics  grDevices utils     datasets  methods   base

other attached packages:
 [1] here_1.0.1         modelsummary_2.3.0 haven_2.5.5        lubridate_1.9.4
 [5] forcats_1.0.0      stringr_1.5.1      dplyr_1.1.4        purrr_1.0.4
 [9] readr_2.1.5        tidyr_1.3.1        tibble_3.2.1       ggplot2_3.5.2
[13] tidyverse_2.0.0

loaded via a namespace (and not attached):
 [1] gtable_0.3.6        xfun_0.52           bayestestR_0.15.3
 [4] insight_1.2.0       lattice_0.22-7      tzdb_0.5.0
 [7] vctrs_0.6.5         tools_4.5.1         generics_0.1.4
[10] parallel_4.5.1      datawizard_1.1.0    sandwich_3.1-1
[13] fansi_1.0.6         pkgconfig_2.0.3     Matrix_1.7-3
[16] tinytable_0.9.0     data.table_1.17.2   checkmate_2.3.2
[19] RColorBrewer_1.1-3  lifecycle_1.0.4     compiler_4.5.1
```

```
[22] farver_2.1.2         codetools_0.2-20     litedown_0.7
[25] htmltools_0.5.8.1    yaml_2.3.10          pillar_1.10.2
[28] MASS_7.3-65          multcomp_1.4-28      parallelly_1.46.0
[31] tidyselect_1.2.1     digest_0.6.37        performance_0.13.0
[34] mvtnorm_1.3-3        stringi_1.8.7        future_1.68.0
[37] listenv_0.10.0       splines_4.5.1        labeling_0.4.3
[40] rprojroot_2.0.4      fastmap_1.2.0        grid_4.5.1
[43] cli_3.6.5            magrittr_2.0.3       survival_3.8-3
[46] TH.data_1.1-3        future.apply_1.20.1  withr_3.0.2
[49] scales_1.4.0         backports_1.5.0      timechange_0.3.0
[52] estimability_1.5.1   rmarkdown_2.29       emmeans_1.11.1
[55] globals_0.18.0       zoo_1.8-14           hms_1.1.3
[58] coda_0.19-4.1        evaluate_1.0.3       knitr_1.50
[61] parameters_0.25.0    lmtest_0.9-40        rlang_1.1.6
[64] xtable_1.8-4         glue_1.8.0           rstudioapi_0.17.1
[67] jsonlite_2.0.0       effectsize_1.0.0     R6_2.6.1
[70] tables_0.9.31
```